

Part 3 - Machine Learning and AI

Final Capstone Report: Intelligent Mobility Solution

Methodology, findings, deployment simulation and responsible-AI assessment

Important accident-data limitation. No real accident dataset was supplied. The classification target `high_risk` is a documented proxy created from High/Severe congestion occurring with severe or low-visibility weather. Classification results demonstrate the machine-learning workflow only and must not be interpreted as prediction of actual accidents.

Executive Summary

Part 3 transforms the cleaned traffic workflow from Part 2 into an intelligent mobility prototype covering supervised and unsupervised learning, deep learning, explainability, experiment tracking, recommendations, deployment simulation, monitoring and responsible AI. The verified input contains 48,187 cleaned hourly observations. Across the tasks, the strongest recurring finding is that temporal patterns - especially hour of day and weekday/weekend structure - are more informative for traffic demand than individual weather variables.

Area	Key verified result
Supervised classification	Random Forest: F1 0.9415; ROC AUC 0.9982 on the proxy <code>high_risk</code> target
Supervised regression	HistGradientBoosting: MAE 257.60 vehicles/hour; R^2 0.9507
Unsupervised learning	6 K-means clusters selected; silhouette 0.4112; 125 association rules; highest lift 3.663
Deep learning	MLP: MAE 282.00 vehicles/hour; R^2 0.9468
Explainability	SHAP ranked cyclical hour and other temporal features as the strongest demand drivers
Recommendation	Lowest weekday historical window: 02:00-03:00, about 302 vehicles/hour
Deployment & monitoring	FastAPI endpoints returned HTTP 200; feature drift triggered ALERT / Requires investigation

Data and Methodological Foundation

The Part 3 workflow uses the cleaned Part 2 dataset rather than the raw source. The final input has 48,187 rows and 9 base columns. Time, weather, holiday and cyclical encodings are generated within the modelling scripts as required. Logging is implemented in each module to record normal milestones, warnings and errors, while trained models and task outputs are saved as reproducible artifacts.

For the classification workflow, congestion categories are recalculated from traffic-volume quartiles ($Q1 = 1,192.5$; $Q2 = 3,379$; $Q3 = 4,933$). The proxy `high_risk` target is 1 only when High/Severe congestion occurs together with severe weather (Fog, Snow, Squall or Thunderstorm) or low visibility. This produced 5,438 positive and 42,749 negative proxy observations. Direct target-derived fields were excluded from the classification predictors to reduce leakage.

Task 1 - Supervised Machine Learning

Methodology. A common engineered feature set was used for classification and regression, including time variables, weather encodings, a holiday flag and cyclical hour/day-of-week representations. Logistic Regression and Random Forest were compared for proxy high-risk classification. Linear Regression and HistGradientBoosting were compared for traffic-volume regression. A fixed 80/20 train-test split with `random_state=42` was used.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	0.9656	0.7747	0.9798	0.8653	0.9935
Random Forest	0.9867	0.9364	0.9467	0.9415	0.9982

Model	MAE (vehicles/hour)	R ²
Linear Regression	818.03	0.7149
HistGradientBoosting	257.60	0.9507

Findings. Random Forest provided the stronger overall classification balance, while Logistic Regression achieved slightly higher recall. HistGradientBoosting substantially outperformed Linear Regression for traffic demand, reducing MAE to about 258 vehicles/hour and explaining about 95.1% of test-set variance. The exceptionally high classification scores must be interpreted cautiously because the target is a deterministic proxy constructed from traffic and weather conditions, not observed accidents.

Task 2 - Unsupervised Machine Learning

Methodology. K-means clustering grouped traffic conditions using cyclical hour information, weather severity and traffic volume. Candidate solutions from k=2 to k=6 were compared using silhouette score. Association-rule mining discretised time period, weekday type, weather and quartile-based congestion level, then ranked rules by lift.

Findings. The six-cluster solution achieved the highest tested silhouette score (0.4112). The clusters separated recognisable operating states: low-volume overnight conditions, moderate late-evening traffic, high-demand morning and midday conditions, and the highest-volume evening peak. The evening-peak cluster averaged about 5,125 vehicles/hour and was dominated by Severe congestion. A separate low-volume overnight/clear cluster averaged about 850 vehicles/hour.

Association-rule mining produced 125 rules, with the highest lift of 3.663. Overnight weekend conditions were strongly associated with Low congestion, while several weekday-morning combinations were associated with Severe congestion. These are co-occurrence patterns and should not be interpreted as causal relationships.

Task 3 - Deep Learning and Explainability

Methodology. A multilayer perceptron (MLPRegressor) was trained for traffic-demand prediction using 23 engineered features. For efficient training, a reproducible sample of 12,000 observations was used, split into 9,600 training and 2,400 test observations after feature scaling. SHAP explainability was applied to a comparable Random Forest regressor trained on the same prediction problem because tree-based SHAP is more direct to interpret than the MLP.

Deep-learning metric	Result
MAE	282.00 vehicles/hour
R ²	0.9468
Training iterations	300 (configured limit reached)

Findings. The neural network achieved strong held-out demand prediction. It reached the configured 300-iteration limit before formal convergence, so further tuning remains possible. SHAP showed that hour_cos was the most influential feature, followed by hour, hour_sin, day-of-week cyclical features and weekend status. Temperature contributed less than the main temporal variables, while individual weather categories had smaller effects. This reinforces the earlier capstone finding that temporal structure is the dominant driver of traffic demand.

Task 4 - Advanced AI Technique: MLflow

Selection and implementation. MLflow was selected because it directly supports the later MLOps requirements and provides a traceable record of model experiments. A local SQLite tracking backend was used. Four Task 1

models were logged as separate runs under the Smart City Traffic Intelligence - Part 3 experiment, with task/model metadata, version tags, evaluation metrics and source model artifacts.

Value. MLflow creates an auditable comparison history and links model versions to their measured performance and artifacts. It also provides a foundation for deployment and monitoring records in Task 6.

Limitation. The implementation is a local academic simulation rather than a shared production tracking server or enterprise model registry; it demonstrates experiment governance but not production-scale access control or infrastructure.

Task 5 - Traffic Recommendation System

Methodology. Because the dataset represents one corridor, the recommendation component focuses on travel timing rather than route selection. Historical observations are filtered by day type and optional weather condition, aggregated by hour, and ranked to identify lower-traffic travel windows. The output is converted into a plain-language recommendation.

Scenario	Recommended window	Historical average
Weekday, any weather	02:00-03:00	~302 vehicles/hour
Weekday, Rain	02:00-03:00	~299 vehicles/hour

Findings. The similarity between the general weekday and rainy-weekday recommendations suggests that time of day is more influential than rain in identifying the lowest-demand periods in these scenarios. The system is based on historical averages, not real-time conditions, and therefore provides decision support rather than a guaranteed forecast. A production version should also allow user constraints, since the lowest-traffic time may not be the most practical travel time.

Task 6 - MLOps and Deployment Simulation

Model versioning and experiment tracking. A version registry documents the four validated Task 1 models and their metrics. The existing MLflow experiment was extended with a deployment/monitoring run so that experimentation, model versions and operational simulation are traceable within one workflow.

Deployment. A FastAPI mock-up serves the HistGradientBoosting traffic-volume model through /health and /predict endpoints. Both returned HTTP 200 in the verified test. For a sample request dated 2017-08-01 16:00, the API returned a predicted traffic volume of 6,432.68 vehicles/hour and the model version task1-v1-hist_gradient_boosting_regressor.

Monitoring and alerting. Chronological baseline/current slices were compared using prediction error and Population Stability Index (PSI) for feature distributions. Baseline MAE was 255.23 and current MAE was 228.19, giving an MAE ratio of 0.894, so error drift did not trigger an alert. However, clouds_all produced PSI = 0.508, above the illustrative 0.2 threshold. The system therefore returned ALERT / Requires investigation and logged a WARNING. This alert indicates input-distribution change requiring review; it does not by itself prove model failure.

Task 7 - Responsible and Sustainable AI

Bias and fairness. Data coverage is uneven across years, weather, holidays and time periods, so aggregate performance may conceal weaker performance in sparse conditions. The strongest limitation is the proxy high_risk target: it represents a rule-based combination of congestion and adverse/low-visibility weather, not actual accident occurrence. High classification performance therefore shows that the models can learn the proxy definition, not that they can predict real accidents. Before operational use, errors should be assessed by time period, weather condition, day type and other relevant subgroups with sample sizes reported.

Governance. Models should remain decision-support tools with human oversight. Real deployment would require documented data provenance, approval of target definitions, model/version review, subgroup

validation, access controls, rollback procedures, incident handling and monitoring thresholds validated against operational evidence. Alerts should trigger investigation rather than automatic operational action.

Sustainability. Linear models are inexpensive to train and serve, while ensembles and neural networks provide stronger predictive performance at higher computational cost. Retraining should therefore be driven by meaningful drift or operational need rather than performed unnecessarily. Where a simpler model provides adequate value, its lower memory, training and inference requirements may be preferable.

Integrated Findings and Smart-City Implications

Across Parts 1-3, the evidence is consistent: traffic demand is strongly structured by time of day and day type, while weather contributes secondary information. Supervised regression, clustering, SHAP and the recommendation system all reinforce the importance of temporal demand patterns. For a smart-city mobility team, this supports time-sensitive traffic management, demand forecasting and traveller guidance, particularly around morning and evening peak periods.

The project also demonstrates why operational controls matter. Strong model metrics are not sufficient on their own: the classification target is only a proxy, data coverage is uneven, and Task 6 detected meaningful feature-distribution drift even when prediction error did not worsen. A reliable mobility AI service therefore needs data-quality controls, explainability, versioning, monitoring, alert review and human governance alongside predictive models.

Limitations and Conclusion

This capstone is an academic prototype based on a single traffic corridor and historical observations. It does not contain real accident outcomes, multi-route network data or real-time feeds. The recommendation engine therefore addresses travel timing rather than route optimisation, and the high_risk classifier must not be used as an accident-risk decision system. The neural network also reached its configured iteration limit, and monitoring thresholds are illustrative rather than production policy.

Despite these constraints, the final solution demonstrates a complete reproducible AI lifecycle: engineered data, supervised and unsupervised learning, deep learning, SHAP explainability, MLflow experiment tracking, a practical recommendation component, FastAPI deployment, drift monitoring, alerting and responsible-AI assessment. The strongest operational conclusion is that temporal features provide the clearest signal for traffic demand, while governance and monitoring are essential before any model is trusted for real-world mobility decisions.

Deliverable Evidence Produced

The Part 3 submission includes trained model artifacts, task scripts, the clean Tasks 1-6 execution notebook, model metrics, clustering and association-rule outputs, SHAP results, figures, MLflow tracking data, recommendation outputs, FastAPI deployment files, monitoring reports/logs, this final capstone report, and the separate Responsible and Sustainable AI report.